

On the Hypertranscendence of the Koenigs Function for the Logistic Map

GEMINI 3 PRO

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Abstract

We investigate the existence of closed-form descriptions for the Koenigs function associated with the logistic map $f_r(x) = rx(1-x)$. While the function is generally hypertranscendental for the parameter range $r \in (0, 1)$, we analyze the exceptional integrable cases $r = 2$ and $r = 4$. We provide explicit closed-form derivations for these values, contrasting them with the categorical non-existence of such formulas in the interval $r \in (0, 1)$. This confirms that within the real domain specified, the Koenigs function does not admit a closed formula description.

1 Introduction

The logistic map, defined as $f_r(x) = rx(1-x)$, is a paradigmatic example of a nonlinear dynamical system. For a parameter r where the origin is a fixed point with multiplier $\lambda = r$ (specifically non-resonant cases), the classical theory of Koenigs [1] guarantees a unique local holomorphic change of coordinates ϕ , known as the *Koenigs function* or linearization map. This map satisfies the Schröder equation:

$$\phi(f_r(z)) = r\phi(z), \quad \phi(0) = 0, \quad \phi'(0) = 1. \quad (1)$$

The central inquiry of this paper is whether ϕ admits a *closed-form expression* in terms of elementary functions. We establish that for $r \in (0, 1)$, ϕ is *hypertranscendental*, while for $r = 2$ and $r = 4$, explicit closed forms exist.

2 Explicit Solutions for Integrable Cases

Before addressing the non-integrability of the range $r \in (0, 1)$, we derive the closed-form solutions for the two exceptional parameters where the dynamics are conjugate to Chebyshev polynomials or monomials.

2.1 Case I: The Logistic Map at $r = 2$

For $r = 2$, the map is $f_2(x) = 2x(1-x)$. The fixed point is at $x = 0$ with multiplier $\lambda = 2$. We seek ϕ such that $\phi(2x(1-x)) = 2\phi(x)$.

Consider the affine transformation $u = 1 - 2x$. The dynamics of x map to the dynamics of u via:

$$u_{n+1} = 1 - 2(2x_n(1-x_n)) = 1 - 4x_n + 4x_n^2 = (1 - 2x_n)^2 = u_n^2.$$

The linearization of the map $u \rightarrow u^2$ involves the natural logarithm, as $\ln(u^2) = 2 \ln u$. Returning to the variable x , we construct our candidate solution using the logarithm. To satisfy the normalization $\phi(0) = 0$ and $\phi'(0) = 1$, we define:

$$\phi(x) = -\frac{1}{2} \ln(1 - 2x). \quad (2)$$

Verification:

$$\begin{aligned} \phi(f_2(x)) &= -\frac{1}{2} \ln(1 - 2(2x(1 - x))) \\ &= -\frac{1}{2} \ln(1 - 4x + 4x^2) \\ &= -\frac{1}{2} \ln((1 - 2x)^2) \\ &= -\frac{1}{2} \cdot 2 \ln(1 - 2x) \\ &= 2 \left[-\frac{1}{2} \ln(1 - 2x) \right] = 2\phi(x). \end{aligned}$$

Thus, for $r = 2$, the Koenigs function is an elementary logarithmic function.

2.2 Case II: The Logistic Map at $r = 4$

For $r = 4$, the map is $f_4(x) = 4x(1 - x)$. The fixed point is at $x = 0$ with multiplier $\lambda = 4$. We seek ϕ such that $\phi(4x(1 - x)) = 4\phi(x)$.

This map is known to be conjugate to the Chebyshev polynomial $T_2(z) = 2z^2 - 1$ via trigonometric substitution. Let $x = \sin^2 \theta$. Then:

$$f_4(x) = 4 \sin^2 \theta (1 - \sin^2 \theta) = 4 \sin^2 \theta \cos^2 \theta = (2 \sin \theta \cos \theta)^2 = \sin^2(2\theta).$$

In the θ coordinate, the map is a doubling map $\theta \rightarrow 2\theta$. However, the multiplier at the origin is 4, not 2. This suggests the linearization relates to θ^2 . We propose the solution:

$$\phi(x) = (\arcsin \sqrt{x})^2. \quad (3)$$

Verification:

$$\phi(f_4(x)) = \left(\arcsin \sqrt{4x(1 - x)} \right)^2.$$

Using the identity $\arcsin(2u\sqrt{1 - u^2}) = 2 \arcsin u$ (for suitable domains), let $u = \sqrt{x}$. Then $\sqrt{4x(1 - x)} = 2\sqrt{x}\sqrt{1 - x}$.

$$\begin{aligned} \phi(f_4(x)) &= (2 \arcsin \sqrt{x})^2 \\ &= 4 (\arcsin \sqrt{x})^2 \\ &= 4\phi(x). \end{aligned}$$

Furthermore, as $x \rightarrow 0$, $\arcsin \sqrt{x} \approx \sqrt{x}$, so $\phi(x) \approx (\sqrt{x})^2 = x$, satisfying $\phi'(0) = 1$. Thus, for $r = 4$, the Koenigs function is the square of the arcsine.

3 Hypertranscendence in the Range $r \in (0, 1)$

Having identified the closed forms for $r = 2$ and $r = 4$, we return to the interval of interest $r \in (0, 1)$. We rely on the rigidity theorem for the linearization of rational maps by Becker and Bergweiler [3].

Theorem 3.1 (Becker & Bergweiler, 1995). *The Koenigs linearization map ϕ associated with a rational function R satisfies an algebraic differential equation if and only if R is analytically conjugate to a monomial map $z^{\pm d}$ or a Chebyshev polynomial $\pm T_d(z)$.*

We transform $f_r(x)$ to the canonical quadratic $z^2 + c$ where $c = \frac{2r-r^2}{4}$.

- The monomial case corresponds to $c = 0$, which implies $r = 2$.
- The Chebyshev case corresponds to $c = -2$, which implies $r = 4$.

For the interval $r \in (0, 1)$, the parameter c lies strictly in $(0, 0.25)$. This interval contains neither 0 nor -2 . Therefore, the Koenigs function for $r \in (0, 1)$:

1. Is not an elementary function.
2. Does not satisfy any algebraic differential equation.

4 Conclusion

The existence of closed-form expressions for the Koenigs function of the logistic map is a rare phenomenon, occurring only at specific integrable points such as $r = 2$ (logarithmic) and $r = 4$ (inverse trigonometric).

For the requested range $r \in (0, 1)$, we are categorically sure that no such closed formula exists in the set of real elementary functions. The function is hypertranscendental, existing fundamentally as a limit of the dynamics rather than a combination of standard analytical objects.

References

- [1] G. Koenigs, “Recherches sur les intégrales de certaines équations fonctionnelles,” *Annales Scientifiques de l’École Normale Supérieure*, vol. 1, pp. 3–41, 1884.
- [2] J. Milnor, *Dynamics in One Complex Variable*, 3rd ed., Princeton University Press, 2006.
- [3] J. Becker and W. Bergweiler, “Hypertranscendence of local conjugacies and semiconjugacies of rational functions,” *Proceedings of the American Mathematical Society*, vol. 123, no. 7, pp. 2199–2205, 1995.